Revision Comparison

A grammar edit beside a multiline equation

LANGUAGE-ONLY / SYNTHETIC EXAMPLE

Task. Correct subject–verb agreement. Preserve the setup, dimensions, regularization condition, equation, and concluding statement. This example uses synthetic text.

C1 / Context / editor summary

The passage introduces a regularized least-squares objective. The parameter dimensions and positive regularization condition remain part of the setup shown in both versions below.

C1 / BEFORE **Anchor: "We minimizes"**

Let $X \in \mathbb{R}^{n \times d}$, $y \in \mathbb{R}^n$, $n \geq 1$, and $\lambda > 0$. The parameter is $\theta \in \mathbb{R}^d$.
We minimizes the following regularized objective:

$$F(\theta) = \frac{1}{2n} \sum_{i=1}^{n} \left(\sum_{j=1}^{d} X_{ij}\theta_j - y_i \right)^2 + \frac{\lambda}{2} \|\theta\|_2^2 \tag{B1}$$
$$= \frac{1}{2n} \|X\theta - y\|_2^2 + \frac{\lambda}{2} \sum_{j=1}^{d} \theta_j^2.$$

The two lines of Equation (B1) express the same objective.

C1 / AFTER **Anchor: "We minimize"**

Let $X \in \mathbb{R}^{n \times d}$, $y \in \mathbb{R}^n$, $n \geq 1$, and $\lambda > 0$. The parameter is $\theta \in \mathbb{R}^d$.
We minimize the following regularized objective:

$$F(\theta) = \frac{1}{2n} \sum_{i=1}^{n} \left(\sum_{j=1}^{d} X_{ij}\theta_j - y_i \right)^2 + \frac{\lambda}{2} \|\theta\|_2^2 \tag{A1}$$
$$= \frac{1}{2n} \|X\theta - y\|_2^2 + \frac{\lambda}{2} \sum_{j=1}^{d} \theta_j^2.$$

The two lines of Equation (A1) express the same objective.

C1 / Reason / editor-proposed

Use "minimize" with "We." Only the verb ending changes; the objective and its conditions stay the same. The changed word is marked in purple.

Meaning and review. Exposition-only. The verb ending is the only source change. The full equation and setup are preserved; this check concerns the text, not a proof of the result.

Versions and coverage. `before.tex` and `after.tex`; hashes in `identity.json`. Entire fragments compared; C1 is the only text edit. **Presentation mapping.** Source `eq:objective` appears as `before:eq:objective` (B1) and `after:eq:objective` (A1), with separate counters and hyperlink destinations. Both source files remain unchanged. With annotations off, both passages and equations remain visible.